Predicting the unexpected: how constituent boundaries shape N400 anticipation and propagation in comprehension and mental production of sentences.

While exploring working-memory effects in sentence recall, Mascelloni et al. (2019) reported that in a word-by-word sentence reading task in German followed by subvocal repetition (**SR)** of the same sentence, an early negative ERP emerged during SR two words before the semantically anomalous item—deviating from the prototypical N400 normally elicited at the anomaly itself (Fig. 1) . Importantly, their materials created the semantic anomaly by switching the verb to produce a clash with the object. The early negativity during repetition was indeed observed on the verb (*The woman **navigates** the shoe*). One possible explanation is an anticipatory effect; another is that the verb was the only word differing between conditions. However, since the verb is the head of the VP, the shift might also reflect a sensitivity of N400 to constituent boundaries. To investigate these possibilities, we designed a set of EEG studies aimed at: **(i)** testing whether the effect generalizes across languages (Italian vs. the original German) and modalities (auditory vs. written), **(ii)** controlling for the verb-change confound, **(iii)** disentangling the anticipatory vs syntactic-sensitivity hypothesis by **(a)** designing sentences in which the constituents onsets and the words preceding the anomaly do not coincide and **(b)** testing the constituent-sensitivity hypothesis with another phenomenon: the ability of an N400 to extend to semantic anomalies that do/don't cross a constituent boundary (DP).

Participants (N = 35) were asked to listen to 200 Italian sentences that could be either correct (*A customer books this **table** for tomorrow*) or semantically anomalous (*A customer books this **enemy** for tomorrow*), then mentally repeat them (SR) at the same speed as the auditory sentence. The object (5th position) was manipulated in the semantically anomalous condition, creating a clash with V in 3rd position and with the subsequent PP. Additionally, half of the ending prepositional phrases (PPs) depended on the object DP (**extDP** condition), the other half functioned as verbal complements (**VPmod**), resulting in 4 conditions: Correct VPmod (e.g., *A customer books [this table] for tomorrow*), Correct ext-DP (e.g., *The postman collects many [letters of congratulations]*), Semantic VPmod (e.g., *A customer books [this enemy] for tomorrow*), Semantic ext-DP (e.g., *The postman collects many [cultures of congratulations]*). This design can now address two questions: **RQ1**, do semantic effects during SR appear at the verb or at the object? **RQ2**, does the expected N400 at the object propagate further, and if so, does such propagation depend on PP attachment?

Results revealed a centroparietal N400 at the keyword and a parietal–occipital LPC for anomalous vs. correct sentences during listening. For **RQ1**, a mild but significant N400-like effect emerged at the verb during SR (Fig. 2), replicating previous findings but leaving open whether the shift reflects anticipation or N400 sensitivity to constituent boundaries. To clarify this, a second study was conducted where anticipation, constituent onset, and verb position are dissociated (Fig. 4); preliminary analyses (N = 22) show no significant effects so far.

For **RQ2**, the LPC propagated across all subsequent words in anomalous sentences, regardless of PP attachment. In contrast, an N400 at the PP noun appeared only in the Semantic extended-DP condition (Fig. 3), suggesting that N400 effects are indeed sensitive to constituent structure and syntactic dependency. Future data from an increased sample of Study 2 will be crucial for determining whether the negativity in SR reflects anticipation or actual sensitivity to constituent structure.

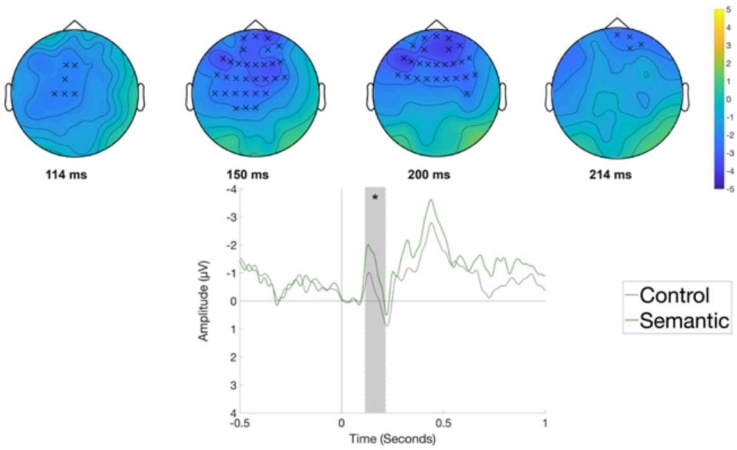

Figure 1. From Mascelloni et al. (2019). Contrast between Correct (*The woman **ties** the shoe*) and Semantic sentences (*The woman **navigates** the shoe*) at verb position (**ties**/**navigates**) in subvocal repetition.

Study 1 - Results

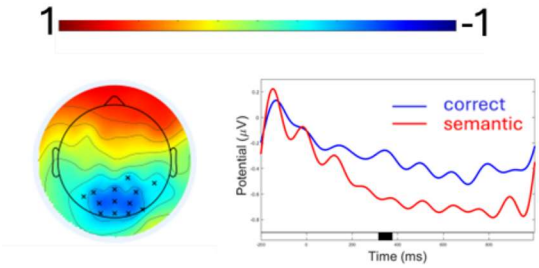

Figure 2. Contrast between Correct (*A customer books this **table** for tomorrow*) and Semantic sentences (*A customer books this **enemy** for tomorrow*) at verb position (***books***) in subvocal repetition.

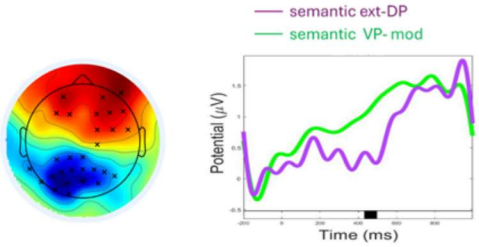

Figure 3. Contrast between extDP (*The postman collects many [**cultures** of congratulations]*) and VPmod (*A customer books [this **enemy**] for tomorrow*) semantically anomalous sentences at the PP noun during listening.

Study 2 - Paradigm

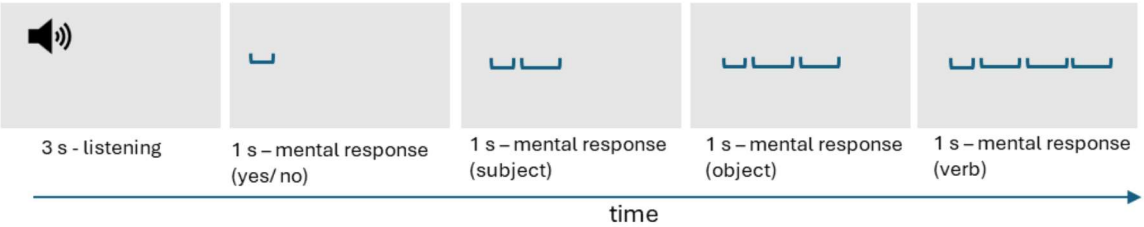

Figure 4. Schematic representation of the experimental paradigm. First, participants listen to a sentence – e.g. *La volpe*[subject] *svuota*[verb] *un **pelo***[object] (the fox empties a hair). They must then *mentally* produce a reply in a fixed structure: *No, la volpe* [subject] *un **pelo*** [object] *non lo svuota* [verb]. They do this by mentally uttering one constituent at a time, synchronized with guiding lines that appear on the screen at a rate of one per second. The subject–object–verb order is a possible Italian structure used in corrective contexts. This structure allows a dissociation between (i) the point where an anticipatory effect would occur (the subject, two words before the semantic anomaly), (ii) the onset of the predicate (object), and (iii) the verb (final position in this structure). The paradigm includes one second break between the presentation of the sentence and the mental response, and an overt repetition of the response followed by second mental repetition

References

Mascelloni, M., Zamparelli, R., Vespignani, F., Gruber, T., Mueller, J. L., & Duff, M. (2019). Distinct Neural Processes for Memorizing Form and Meaning Within Sentences. *Frontiers in Human Neuroscience*, *13*, 1–18. https://doi.org/10.3389/fnhum.2019.00412